

Connecting to the internship database

Data Science & Analytics Internship · a complete walkthrough · allow 20 minutes

This guide takes you from nothing installed to running your first query. It assumes you have never used a database before. If you have, skip to Part 2.

Everyone does this on their own machine. Not one person for the group. A connection that works on your group lead's laptop tells you nothing about yours, and week 2 starts with all of you querying at once.

What you are connecting to

The internship data lives on a PostgreSQL database server running on Amazon's infrastructure. It is not on your laptop. You connect to it over the internet, run queries, and get results back. Every group connects to the same server, but each group sees only its own data.

The words you will keep seeing

These come up constantly. Five minutes here saves confusion later.

Host	The address of the server, like a web address. Ours is a long one ending in <code>rds.amazonaws.com</code> .
Port	Which door on that server to knock on. PostgreSQL uses 5432. You will never need to change it.
Database	A container holding many tables. Ours is called <code>internship</code> .
Schema	A folder inside the database that groups related tables together. You have two: one to read from, one to write to.
Table	Rows and columns, like a spreadsheet tab. Your data is spread across four of them.
Username / role	Your group name — <code>group1a</code> , <code>group2b</code> and so on. It controls what you are allowed to see and change.
SSL	Encryption for the connection, so nobody between you and the server can read what passes. Always on.
Client	The program on your computer that talks to the server. DBeaver is one. This guide uses it.

Your two schemas

This is the part people find confusing at first, so it is worth being clear about now.

Your raw schema — named `raw_retail`, `raw_clinic`, `raw_school` or `raw_rides` depending on your domain — holds the original data. You can read it. You cannot change it. Neither can anyone else, which is what guarantees you and your paired group started from exactly the same place.

Your group schema — named after your group, like `group1a` — is empty right now. From week 3 this is where your cleaned tables go. You can do anything you like in here. Nobody outside your group can see it, including the group you are paired with.

internship	the database
<code>raw_retail</code>	<code>read only</code> - the original data
<code>sales, customers, products, stores</code>	
<code>group1a</code>	<code>yours</code> - empty until week 3

Replace `raw_retail` and `group1a` with your own names throughout this guide. Both are on your access check sheet.

Part 1 — Installing DBeaver

DBeaver is a free program for working with databases. There are other options and they all work; we recommend DBeaver because it runs on every operating system and needs no configuration beyond the connection itself.

1 Download it

Go to **dbeaver.io** and click Download. You want **DBeaver Community Edition**, which is the free one. Ignore DBeaver PRO — it is paid and you do not need anything in it.

System	Which file	Notes
Windows	Windows (installer) 64 bit	Run the .exe and accept the defaults. If Windows shows a security warning, choose More info then Run anyway.
Mac	MacOS dmg (Intel or Apple Silicon)	Open the .dmg and drag DBeaver into Applications. Check which chip you have under Apple menu > About This Mac.
Linux	Debian package or Flatpak	On Ubuntu: <code>sudo dpkg -i</code> the downloaded .deb file.

The download is around 130 MB. On a slow connection, start it before you read the rest of this page.

2 First launch

The first time DBeaver opens it takes a while and may show a progress bar at the bottom. That is normal — it is setting up a workspace folder.

You may be asked whether to create a sample SQLite database. Say **no**. It has nothing to do with your project and will only clutter the sidebar.

You may also be asked about sending anonymous usage statistics. Either answer is fine.

What you should see

A mostly empty window. On the left is the **Database Navigator** — an empty panel where your connection will appear. Along the top is a menu bar with File, Edit, Navigate, **SQL Editor**, **Database**, Window and Help. Those two bold ones are the only menus you will use.

Before you go further, have these three things to hand: your access check sheet (it has the host, your username and your schema names), your group password from your group lead, and a working internet connection. Steps 5 and 6 both need internet.

Part 2 — Creating the connection

You do this once. After it is saved, opening DBeaver takes you straight to your data.

3 Open the new connection wizard

From the menu bar choose **Database > New Database Connection**. A window opens showing a grid of database logos — dozens of them.

Find and select **PostgreSQL**, the blue elephant. There is a search box at the top if the grid is overwhelming. Click **Next**.

Several entries look similar — Greenplum, Redshift, CockroachDB, Timescale — because they are all built on PostgreSQL. Pick plain **PostgreSQL**. The others will fail in ways that are hard to diagnose.

4 Fill in the Main tab

This is the only screen where you type anything. It has tabs across the top — Main, PostgreSQL, Driver properties, SSH, Proxy, **SSL**. Start on Main.

Server / Host	internship-db.coh86gwewtxb.us-east-1.rds.amazonaws.com	Exactly as written. No http, no www, no trailing slash.
Port	5432	Already filled in. Leave it.
Database	internship	All lowercase.
Authentication	Database Native	The default. Do not change it.
Username	your group name	Lowercase, e.g. group3a. Not your own name.
Password	from your group lead	Type it by hand. See the warning below.
Save password	tick the box	Otherwise you retype it every session.

Type the password rather than pasting it. Copying from WhatsApp or email very often picks up an invisible space at the end. The error you get back only says authentication failed — it does not mention the space, and people lose an hour to this.

5 Set SSL

Click the **SSL** tab at the top of the same window. Tick **Use SSL**, and set **SSL mode** to **require**. Leave every other field on that tab empty — you do not need certificate files.

This encrypts everything between your laptop and the server, including your password.

6 Download the driver

The first time you connect to a PostgreSQL database, DBeaver needs driver files it does not ship with. A box appears offering to download them. Click **Download** and wait — it is a few megabytes.

This happens once, ever. If your internet drops during it, DBeaver will offer again next time.

Part 3 — Testing and finding your data

7 Test Connection

At the bottom left of the window, click **Test Connection**. Wait a few seconds.

Success looks like a small box saying **Connected**, along with the server version — PostgreSQL 17.5 — and the driver name. Click OK, then **Finish**.

If it fails, do not start changing settings at random. Go to the troubleshooting table in Part 5, find your exact error message, and do what it says.

8 Rename your connection

Your connection appears in the left panel with an unhelpful automatic name. Right-click it, choose **Rename**, and call it something like **Internship — group1a**. Small thing, but useful once you have other connections.

9 Find your schemas

Click the arrow next to your connection to expand it, then keep expanding:

```
Internship - group1a
  Databases
    internship
      Schemas
        group1a          <- yours, empty for now
        raw_retail       <- your data
      Tables
        customers
        products
        sales
        stores
```

You will see only your own two schemas. Other groups' schemas exist on the same server but are invisible to you — including your paired group's. That is deliberate.

Looking at a table

Double-click any table to open it. You get tabs across the top of the panel that appears:

- **Properties** — the columns and their data types
- **Data** — the actual rows — this is the one you want most often
- **ER Diagram** — a picture of how tables relate. Mostly empty here, because the raw data has no foreign keys defined. Finding those relationships is your job.

Browsing the Data tab is fine for a quick look, but it is not analysis. Everything you hand in comes from SQL you wrote and committed, not from clicking around. Get into the SQL editor early.

Part 4 — Running your first queries

10 Open a SQL editor

Make sure your connection is selected in the left panel, then choose **SQL Editor > New SQL Script**. A blank text area opens. This is where you write queries.

How to run things

Action	Windows / Linux	Mac
Run the query your cursor is on	Ctrl + Enter	Cmd + Enter
Run every query in the script	Alt + X	Option + X
Format messy SQL neatly	Ctrl + Shift + F	Cmd + Shift + F
Autocomplete a table name	Ctrl + Space	Ctrl + Space

Queries are separated by semicolons. Ctrl+Enter runs only the one your cursor sits in, which is what you want most of the time.

11 Check you can read

```
SELECT count(*) FROM raw_retail.sales;
```

Substitute your own schema and fact table. The number you should get back is on your access check sheet. If you get it, your read access works.

12 Check you can write

From week 3 you will be writing cleaned tables into your own schema. Confirm now that you can, rather than discovering a problem then.

```
CREATE TABLE group1a.access_test AS SELECT 1 AS it_works;
SELECT * FROM group1a.access_test;
DROP TABLE group1a.access_test;
```

Run each line with Ctrl+Enter, in order. A table is created, returns 1, then is removed.

13 Confirm you are properly isolated

This one is **supposed to fail**. It proves your paired group cannot reach into your work, and you cannot reach into theirs.

```
CREATE TABLE group1b.test (x int);
```

Substitute your paired group's name. You should get **permission denied for schema group1b**. If it succeeds instead, stop and tell the programme lead immediately — something is misconfigured.

Saving your work

SQL scripts in DBeaver are files. Save them with Ctrl+S, give them meaningful names, and keep them in your repository's sql/ folder. A query you ran once and lost is a query you will write again.

Part 5 — When it does not work

Find your exact error message. Try the fix. Change one thing at a time — changing five settings at once means you will never know which one mattered. Still stuck after that? Post in the shared channel, not a private message, and include the **exact error text** — not "it doesn't work". Someone else has almost certainly hit the same thing.

Error	What to do
Connection timed out / it just spins	Almost always internet. Check a website loads. If your connection is fine and it still times out, message the programme lead — it may be a server-side issue affecting everyone.
password authentication failed for user "..."	Retype the password by hand rather than pasting. Check your username is your group name in lowercase with no spaces — group1a, not Group1A or your own name.
FATAL: database "internshp" does not exist	A typo in the Database field. It is internship, all lowercase.
could not translate host name	A typo in the Host field. It must end in rds.amazonaws.com with no http:// in front.
no pg_hba.conf entry ... no encryption	SSL is not switched on. Go back to the SSL tab, tick Use SSL, set mode to require.
Driver download failed	Internet dropped mid-download. Try again. If your network blocks it, try a phone hotspot for that one step.
Connected, but the Schemas folder is empty	Right-click the connection and choose Refresh. If still empty, you are probably signed in with the wrong username.
relation "sales" does not exist	You forgot the schema name. Write raw_retail.sales, not sales.
permission denied for schema raw_...	You are trying to write to the raw data. It is read only. Write to your own group schema instead.
permission denied for schema group...	You are trying to reach another group's schema. Expected — that is the isolation working.

If you prefer a different client

DBeaver is a recommendation, not a requirement. The connection details are identical whatever you use.

- **pgAdmin** — the official PostgreSQL tool. Heavier, runs in a browser tab. Register > Server, then the same host, port, database, username and password. Set SSL mode to require under the Parameters tab.
- **psql** — the command line client. Fastest if you are comfortable in a terminal:

```
psql "host=internship-db.coh86gwevtxb.us-east-1.rds.amazonaws.com port=5432 dbname=internship user=group1a sslmode=require"
```

- **Python in Colab** — you will need this from week 3 anyway. Your week 3 materials cover it.

Four rules

- **Never share your group password outside your group.** Not in the general channel, not with your paired group, not in a screenshot.
- **Never commit the password to GitHub.** It stays in the history even after you delete it.
- **Never try to write to the raw schema.** It is protected anyway.
- **Do not go looking in other groups' schemas.** You cannot get in, and trying is not a good look.

Once all three checks in Part 4 pass, you are done. Nothing else in the internship requires this setup again — from here it is analysis.